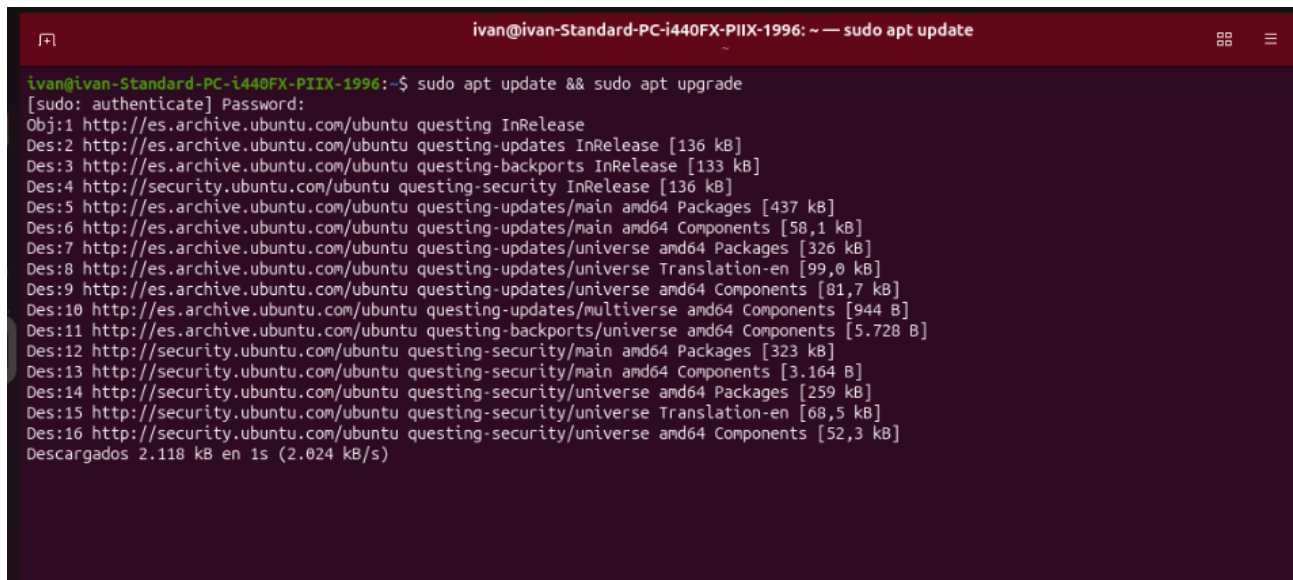


Actividad 1 - Instalación de Docker

Paso 1: Actualizar el sistema

Para llevar a cabo una correcta instalación, primero actualizamos el sistema con:

```
sudo apt update && sudo apt upgrade
```

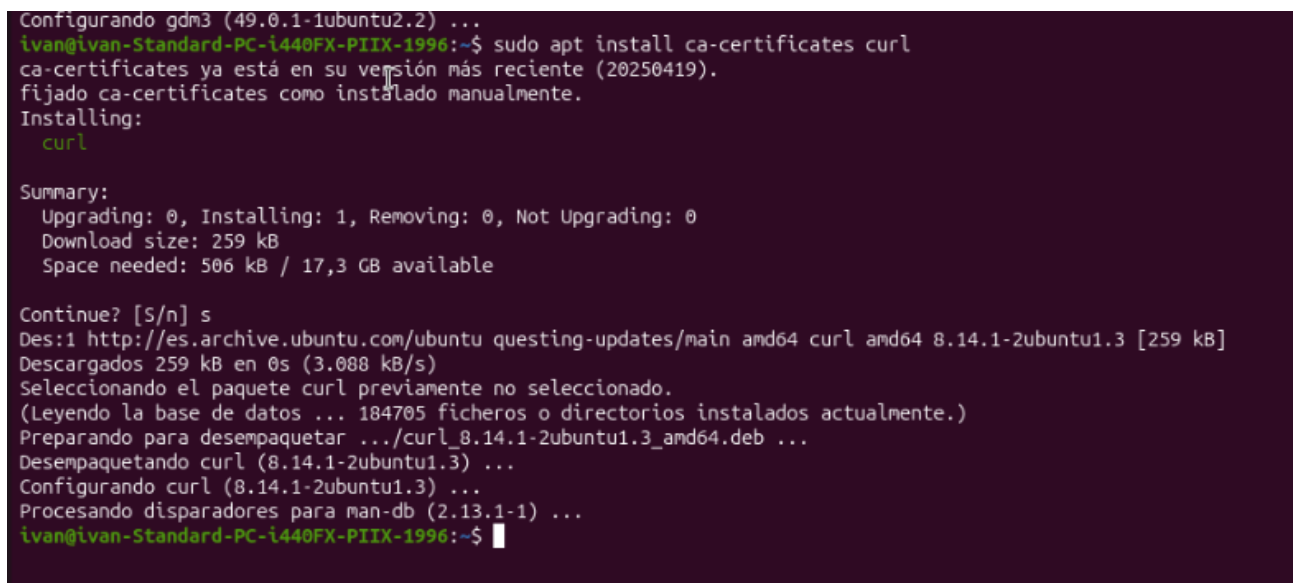
A terminal window titled 'ivan@ivan-Standard-PC-i440FX-PIIX-1996: ~ — sudo apt update'. The terminal shows the output of the command 'sudo apt update && sudo apt upgrade'. It lists various updates from Ubuntu's 'questing' and 'security' repositories, including InRelease files, Packages, and Components. The total download size is 2.118 kB in 1s at 2.024 kB/s.

```
ivan@ivan-Standard-PC-i440FX-PIIX-1996:~$ sudo apt update && sudo apt upgrade
[sudo: authenticate] Password:
Obj:1 http://es.archive.ubuntu.com/ubuntu questing InRelease
Des:2 http://es.archive.ubuntu.com/ubuntu questing-updates InRelease [136 kB]
Des:3 http://es.archive.ubuntu.com/ubuntu questing-backports InRelease [133 kB]
Des:4 http://security.ubuntu.com/ubuntu questing-security InRelease [136 kB]
Des:5 http://es.archive.ubuntu.com/ubuntu questing-updates/main amd64 Packages [437 kB]
Des:6 http://es.archive.ubuntu.com/ubuntu questing-updates/main amd64 Components [58,1 kB]
Des:7 http://es.archive.ubuntu.com/ubuntu questing-updates/universe amd64 Packages [326 kB]
Des:8 http://es.archive.ubuntu.com/ubuntu questing-updates/universe Translation-en [99,0 kB]
Des:9 http://es.archive.ubuntu.com/ubuntu questing-updates/universe amd64 Components [81,7 kB]
Des:10 http://es.archive.ubuntu.com/ubuntu questing-updates/multiverse amd64 Components [944 B]
Des:11 http://es.archive.ubuntu.com/ubuntu questing-backports/universe amd64 Components [5.728 B]
Des:12 http://security.ubuntu.com/ubuntu questing-security/main amd64 Packages [323 kB]
Des:13 http://security.ubuntu.com/ubuntu questing-security/main amd64 Components [3.164 B]
Des:14 http://security.ubuntu.com/ubuntu questing-security/universe amd64 Packages [259 kB]
Des:15 http://security.ubuntu.com/ubuntu questing-security/universe Translation-en [68,5 kB]
Des:16 http://security.ubuntu.com/ubuntu questing-security/universe amd64 Components [52,3 kB]
Descargados 2.118 kB en 1s (2.024 kB/s)
```

Paso 2: Instalar ca-certificates y curl

Instalamos curl para poder descargarnos paquetes a través de una URL:

```
sudo apt install ca-certificates curl
```

A terminal window showing the output of 'sudo apt install ca-certificates curl'. It indicates that 'ca-certificates' is already installed and 'curl' is being installed. It shows the download size (259 kB) and space needed (506 kB). The installation of curl 8.14.1-2ubuntu1.3 is shown, including the preparation of the package and the configuration of the system.

```
Configurando gdm3 (49.0.1-1ubuntu2.2) ...
ivan@ivan-Standard-PC-i440FX-PIIX-1996:~$ sudo apt install ca-certificates curl
ca-certificates ya está en su versión más reciente (20250419).
fijado ca-certificates como instalado manualmente.
Installing:
  curl

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 259 kB
  Space needed: 506 kB / 17,3 GB available

Continue? [S/n] s
Des:1 http://es.archive.ubuntu.com/ubuntu questing-updates/main amd64 curl amd64 8.14.1-2ubuntu1.3 [259 kB]
Descargados 259 kB en 0s (3.088 kB/s)
Seleccionando el paquete curl previamente no seleccionado.
(Leyendo la base de datos ... 184705 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../curl_8.14.1-2ubuntu1.3_amd64.deb ...
Desempaquetando curl (8.14.1-2ubuntu1.3) ...
Configurando curl (8.14.1-2ubuntu1.3) ...
Procesando disparadores para man-db (2.13.1-1) ...
ivan@ivan-Standard-PC-i440FX-PIIX-1996:~$
```

Paso 3: Configurar la clave GPG de Docker

Creamos la carpeta /etc/apt/keyrings, descargamos el paquete de Docker y ajustamos los permisos:

```
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
```

```
Procesando disparadores para man-db (2.13.1-1) ...
ivan@ivan-Standard-PC-i440FX-PIIX-1996:~$ sudo install -m 0755 -d /etc/apt/keyrings
ivan@ivan-Standard-PC-i440FX-PIIX-1996:~$ sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
ivan@ivan-Standard-PC-i440FX-PIIX-1996:~$ sudo chmod a+r /etc/apt/keyrings/docker.asc
ivan@ivan-Standard-PC-i440FX-PIIX-1996:~$ sudo install -m 0755 -d /etc/apt/keyrings
ivan@ivan-Standard-PC-i440FX-PIIX-1996:~$ sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
ivan@ivan-Standard-PC-i440FX-PIIX-1996:~$ sudo chmod a+r /etc/apt/keyrings/docker.asc
ivan@ivan-Standard-PC-i440FX-PIIX-1996:~$
```

Paso 4: Añadir el repositorio Docker a apt

Añadimos el repositorio oficial de Docker a las fuentes de apt:

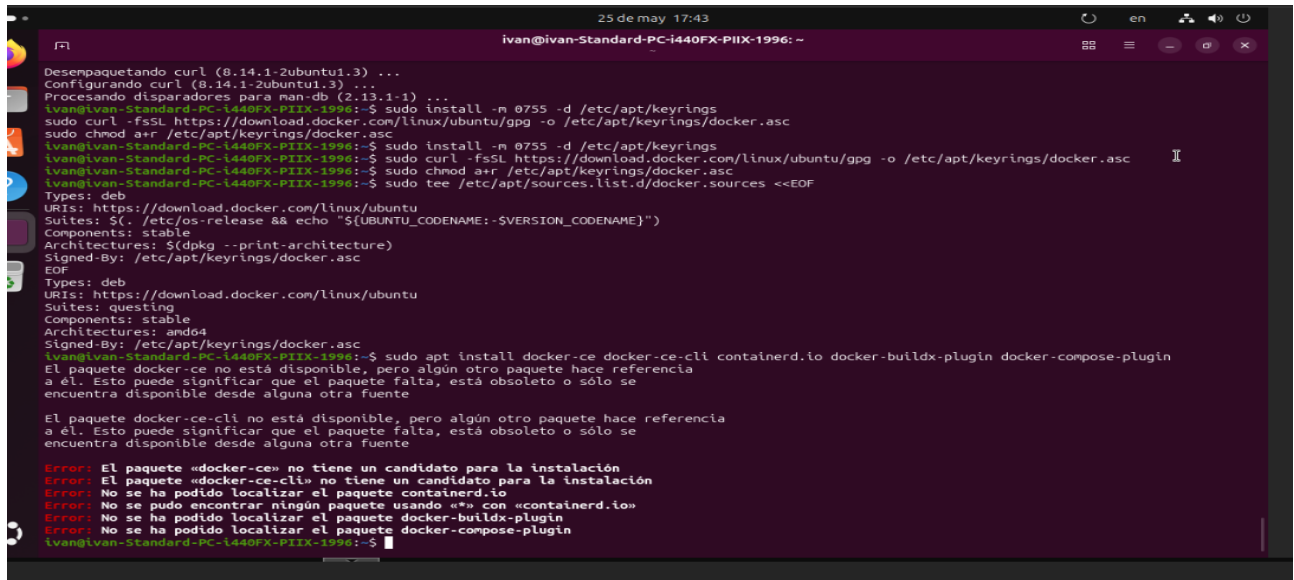
```
sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo ...)
Components: stable
EOF
```

```
ivan@ivan-Standard-PC-i440FX-PIIX-1996:~$ sudo chmod a+r /etc/apt/keyrings/docker.asc
ivan@ivan-Standard-PC-i440FX-PIIX-1996:~$ sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Architectures: $(dpkg --print-architecture)
Signed-By: /etc/apt/keyrings/docker.asc
EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: questing
Components: stable
Architectures: amd64
Signed-By: /etc/apt/keyrings/docker.asc
ivan@ivan-Standard-PC-i440FX-PIIX-1996:~$
```

Paso 5: Instalar Docker

Instalamos Docker y sus componentes. En el primer intento apareció un error porque el repositorio no estaba bien configurado. Tras corregirlo con `sudo apt update` y ajustar el comando de instalación, los paquetes se encontraron correctamente:

```
sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin
docker-compose-plugin
```



A terminal window titled 'ivan@ivan-Standard-PC-i440FX-PIIX-1996: ~' showing the output of a Docker installation command. The terminal displays the progress of installing curl, configuring it, and processing package sources. It then shows the installation of Docker packages. However, several error messages appear: 'El paquete docker-ce no está disponible, pero algún otro paquete hace referencia a él.', 'El paquete docker-ce-cli no está disponible, pero algún otro paquete hace referencia a él.', and 'El paquete «docker-ce» no tiene un candidato para la instalación'. The terminal ends with a prompt for the user to press a key.



A terminal window titled 'ivan@ivan: /' showing the output of the same Docker installation command after repository configuration. The terminal displays the progress of installing curl, configuring it, and processing package sources. It then shows the installation of Docker packages. The output indicates that the packages were installed successfully: 'Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios.' followed by a list of installed packages: 'libgl1-amd-gpu-dri libglapi-mesa libllvm17t64 libwoff1 python3-netifaces'. It also lists additional packages to be installed: 'docker-ce-rootless-extras git git-man liberror-perl libslirp0 pigz slirp4netns'. The terminal ends with a prompt for the user to press a key.

Paso 6: Verificar la instalación

Comprobamos que Docker está activo y en ejecución con:

```
sudo systemctl status docker
```

```
ivan-cliente@ivan-cliente: /
ivan-cliente@ivan-cliente:/$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-05-12 21:33:30 CEST; 1min 51s ago
 TriggeredBy: ● docker.socket
    Docs: https://docs.docker.com
   Main PID: 31906 (dockerd)
     Tasks: 13
    Memory: 25.2M (peak: 26.7M)
       CPU: 1.432s
    CGroup: /system.slice/docker.service
            └─31906 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

May 12 21:53:29 ivan-cliente dockerd[31906]: time="2026-05-12T21:53:29.355230802+02:00" level=info msg="Re:
May 12 21:53:29 ivan-cliente dockerd[31906]: time="2026-05-12T21:53:29.501277217+02:00" level=info msg="De
May 12 21:53:29 ivan-cliente dockerd[31906]: time="2026-05-12T21:53:29.521062534+02:00" level=info msg="De
May 12 21:53:30 ivan-cliente dockerd[31906]: time="2026-05-12T21:53:30.846824464+02:00" level=info msg="Lo
May 12 21:53:30 ivan-cliente dockerd[31906]: time="2026-05-12T21:53:30.873748012+02:00" level=info msg="Do
May 12 21:53:30 ivan-cliente dockerd[31906]: time="2026-05-12T21:53:30.874135768+02:00" level=info msg="In
May 12 21:53:30 ivan-cliente dockerd[31906]: time="2026-05-12T21:53:30.938102863+02:00" level=info msg="Co
May 12 21:53:30 ivan-cliente dockerd[31906]: time="2026-05-12T21:53:30.968437717+02:00" level=info msg="Da
May 12 21:53:30 ivan-cliente dockerd[31906]: time="2026-05-12T21:53:30.968751260+02:00" level=info msg="AP
May 12 21:53:30 ivan-cliente systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-22/22 (END)
```